

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

November 2023 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

36 adaptive question sections 113 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. A source grid, graph, table, or diagram is retained once at a usable size.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 2; the selection contains 113 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

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Question 01

4MA1-1H Q3 demand

MODULAR UNIT 2

4 marks

3 Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

.....%

WORKING SPACE

Question 02

4MA1-1H Q6 demand

MODULAR UNIT 2

4 marks

- 6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket **A**
 $\frac{1}{6}$ off normal price

ticket **B**
20% off normal price

The sale price of ticket **A** is 140 euros.

The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

WORKING SPACE

Question 03

4MA1-1H Q7 demand

MODULAR UNIT 2

2 marks

7 $A = 5^3 \times 7^3 \times 11^6$ and $B = 5^6 \times 7^2 \times 11^4$

Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

.....

WORKING SPACE

Question 04

4MA1-1H Q8 (a)

MODULAR UNIT 2

2 marks

- 8 (a) Solve the inequality $8x - 4 \geq 3x - 10$

.....
(2)

WORKING SPACE

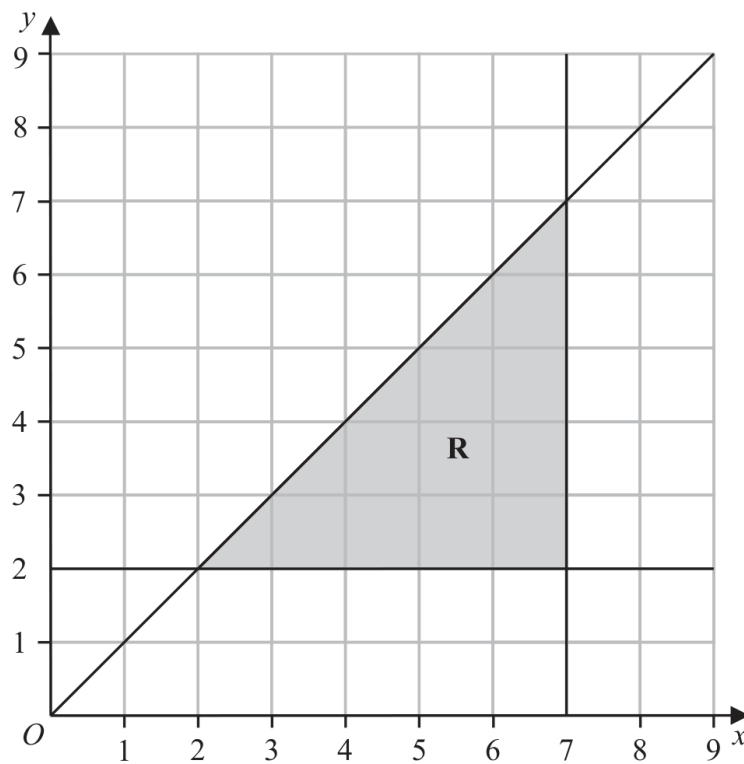
Question 05

4MA1-1H Q8 (b)

MODULAR UNIT 2

3 marks

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down the three inequalities that define the region **R**

.....
.....
.....
(3)

WORKING SPACE

Working space for Question 05

Continue your method

UNIT 2**3 marks**

WORKING SPACE

Question 06

4MA1-1H Q9 (a)

MODULAR UNIT 2

1 marks

- 9 (a) Write 5.87×10^{-4} as an ordinary number.

.....
(1)

WORKING SPACE

Question 07

4MA1-1H Q9 (b)

MODULAR UNIT 2

1 marks

(b) Write 84 000 000 in standard form.

.....
(1)

WORKING SPACE

Question 08

4MA1-1H Q9 (c)

MODULAR UNIT 2

2 marks

The number of neurons in a human brain is 8.5×10^{10}
The number of neurons in a monkey brain is 1.47×10^9

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

- (c) Work out the value of K
Give your answer correct to one decimal place.

$K =$
(2)

WORKING SPACE

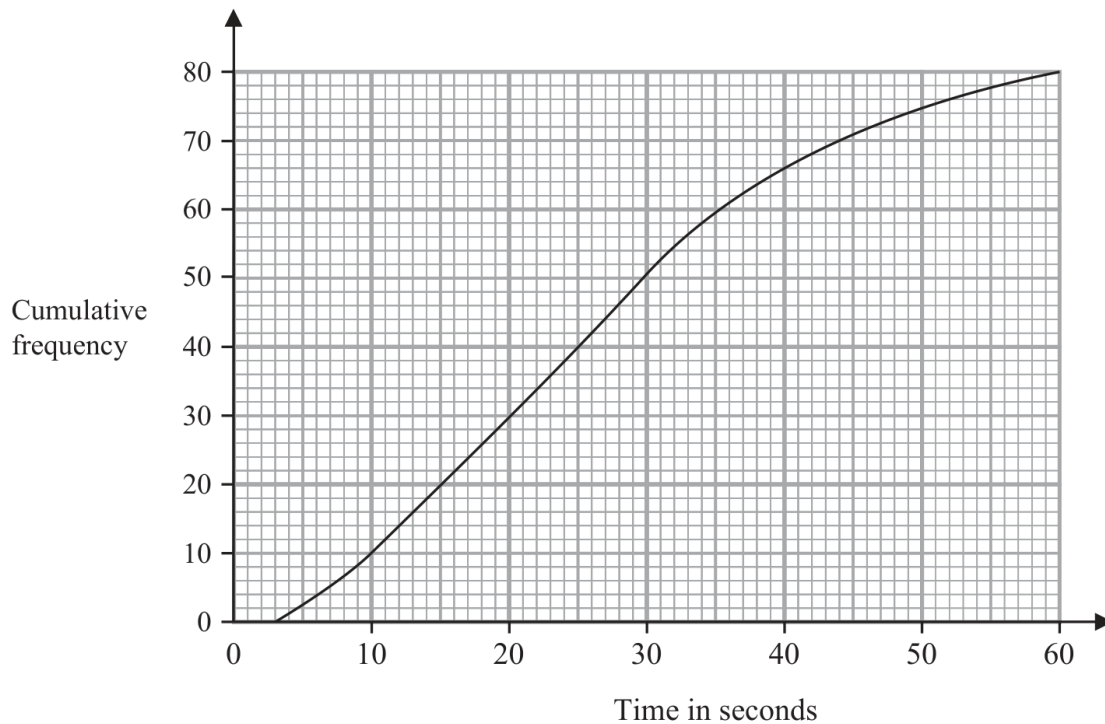
Question 09

4MA1-1H Q13 (a)

MODULAR UNIT 2

1 marks

- 13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (a) Find an estimate for the median time.

..... seconds

(1)

WORKING SPACE

Working space for Question 09

Continue your method

UNIT 2**1 marks**

WORKING SPACE

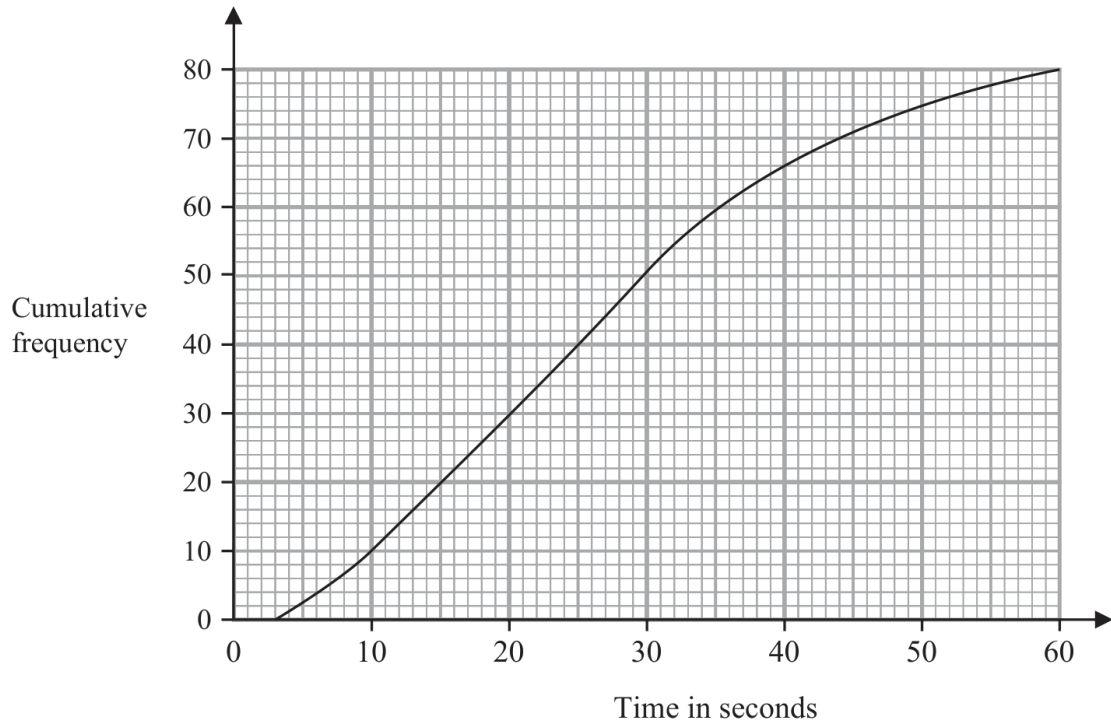
Question 10

4MA1-1H Q13 (b)

MODULAR UNIT 2

3 marks

- 13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

..... %
(3)

WORKING SPACE

Working space for Question 10

Continue your method

UNIT 2

3 marks

WORKING SPACE

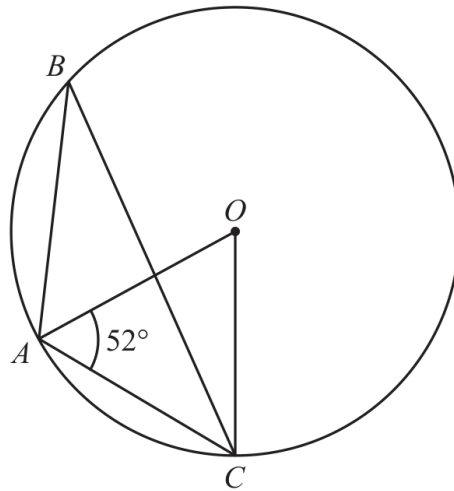
Question 11

4MA1-1H Q14 demand

MODULAR UNIT 2

3 marks

14

Diagram **NOT**
accurately drawn

A , B and C are points on a circle, centre O

Angle $OAC = 52^\circ$

Find the size of angle ABC

Give reasons for your working.

○

WORKING SPACE

Working space for Question 11

Continue your method

UNIT 2

3 marks

WORKING SPACE

Question 12

4MA1-1H Q15 demand

MODULAR UNIT 2

3 marks

15 Make n the subject of the formula $x = \frac{3p+n}{3n-4}$

WORKING SPACE

Question 13

4MA1-1H Q16 demand

MODULAR UNIT 2

4 marks

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

.....

WORKING SPACE

Question 14

4MA1-1H Q18 demand

MODULAR UNIT 2

5 marks

18 The diagram shows the positions of three villages, A , B and C

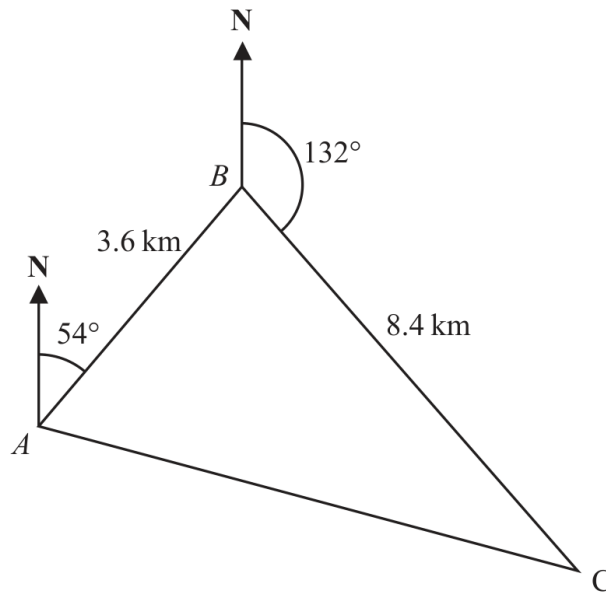


Diagram **NOT**
accurately drawn

The bearing of B from A is 054°

The bearing of C from B is 132°

Melur walks from A to B

She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.

Give your answer in hours and minutes.

..... hours minutes

WORKING SPACE

Working space for Question 14

Continue your method

UNIT 2**5 marks**

WORKING SPACE

Question 15

4MA1-1H Q19 demand

MODULAR UNIT 2

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

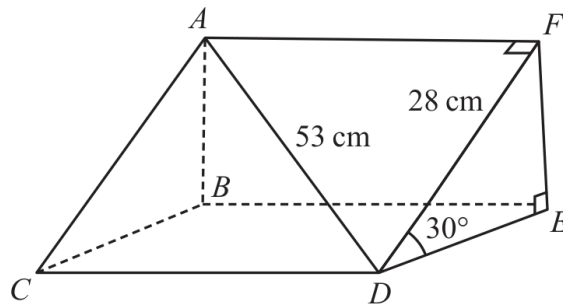
WORKING SPACE

Question 16

4MA1-1H Q21 demand

MODULAR UNIT 2

5 marks

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

WORKING SPACE

Question 17

4MA1-1H Q22 demand

MODULAR UNIT 2

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

Question 18

4MA1-1H Q23 demand

MODULAR UNIT 2

5 marks

23 Solid **A** is similar to solid **B**Here is some information about solid **A** and solid **B**

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid **B**
 Give your answer as a decimal.

..... cm

WORKING SPACE

Question 19

4MA1-1H Q24 (a)

MODULAR UNIT 2

2 marks

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

(a) Write down the value of a and the value of b

 $a = \dots\dots\dots$ $b = \dots\dots\dots$

(2)

WORKING SPACE

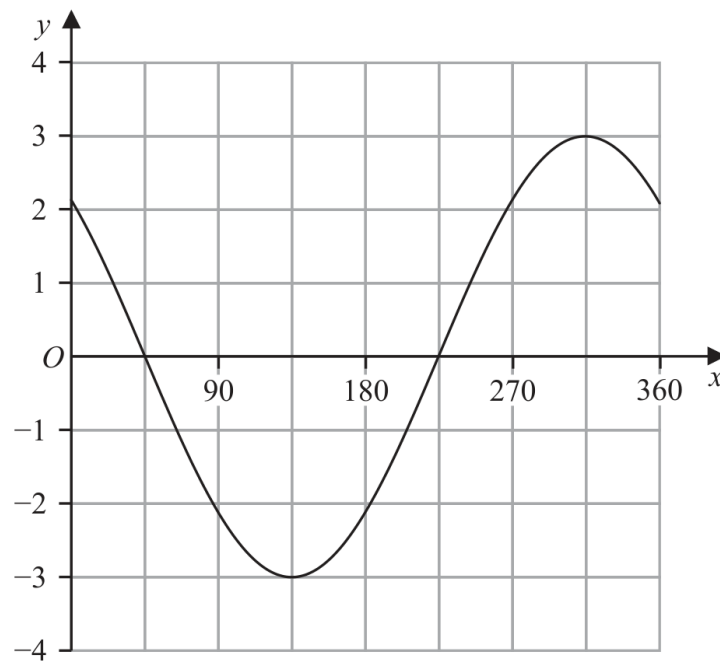
Question 20

4MA1-1H Q24 (b)

MODULAR UNIT 2

2 marks

The graph of $y = p \cos(x + q)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(2)

WORKING SPACE

Working space for Question 20

Continue your method

UNIT 2

2 marks

WORKING SPACE

Question 21

4MA1-2H Q1 (a)

MODULAR UNIT 2

1 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (a) Write down the modal class.

.....
(1)

WORKING SPACE

Working space for Question 21

Continue your method

UNIT 2

1 marks

WORKING SPACE

Question 22

4MA1-2H Q1 (b)

MODULAR UNIT 2

3 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (b) Work out an estimate for the total length, in minutes, of these telephone calls.

..... minutes
(3)

WORKING SPACE

Working space for Question 22

Continue your method

UNIT 2

3 marks

WORKING SPACE

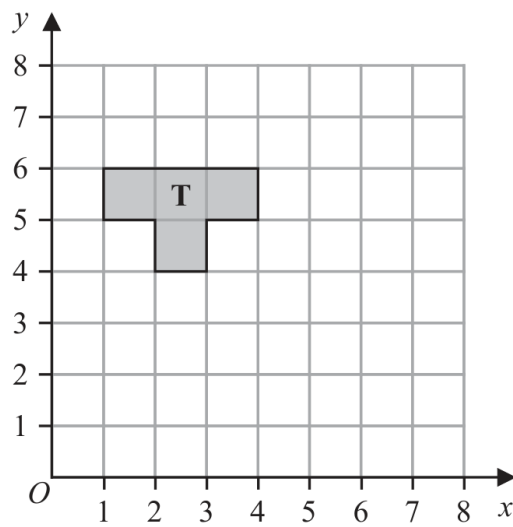
Question 23

4MA1-2H Q3 (a)

MODULAR UNIT 2

2 marks

3

(a) Reflect shape **T** in the line $y = x$

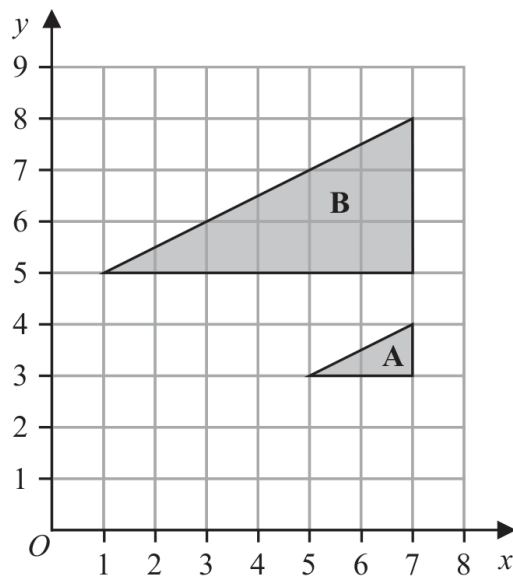
(2)

Question 24

4MA1-2H Q3 (b)

MODULAR UNIT 2

3 marks



(b) Describe fully the single transformation that maps triangle A onto triangle B

(3)

WORKING SPACE

Working space for Question 24

Continue your method

UNIT 2

3 marks

WORKING SPACE

Question 25

4MA1-2H Q5 whole

MODULAR UNIT 2

3 marks

5 Avril bakes a cake.

She uses flour, butter and sugar such that

$$\text{weight of flour : weight of butter} = 6 : 5$$
$$\text{weight of butter : weight of sugar} = 3 : 2$$

Avril uses 120 grams of sugar.

Work out the weight of flour Avril uses.

..... grams

WORKING SPACE

Question 26

4MA1-2H Q7 whole

MODULAR UNIT 2

3 marks

- 7 Hermione buys a boat for \$26 800
The value of the boat depreciates by 8% each year.

Work out the value of the boat at the end of 3 years.
Give your answer correct to the nearest dollar.

\$.....

WORKING SPACE

Question 27

4MA1-2H Q8 whole

MODULAR UNIT 2

3 marks

- 8 The mean number of goals scored by a hockey team in 8 matches is 6
The team plays 2 more matches and scores k goals in each match.
The mean number of goals scored by the hockey team in the 10 matches is 7

Work out the value of k

$k =$

WORKING SPACE

Question 28

4MA1-2H Q13 whole

MODULAR UNIT 2

2 marks

13 Robert asked 11 people how many meetings they attended last week.

Here are the results in numerical order.

1 2 4 6 6 8 11 12 13 14 17

Find the interquartile range of the number of meetings.

.....

WORKING SPACE

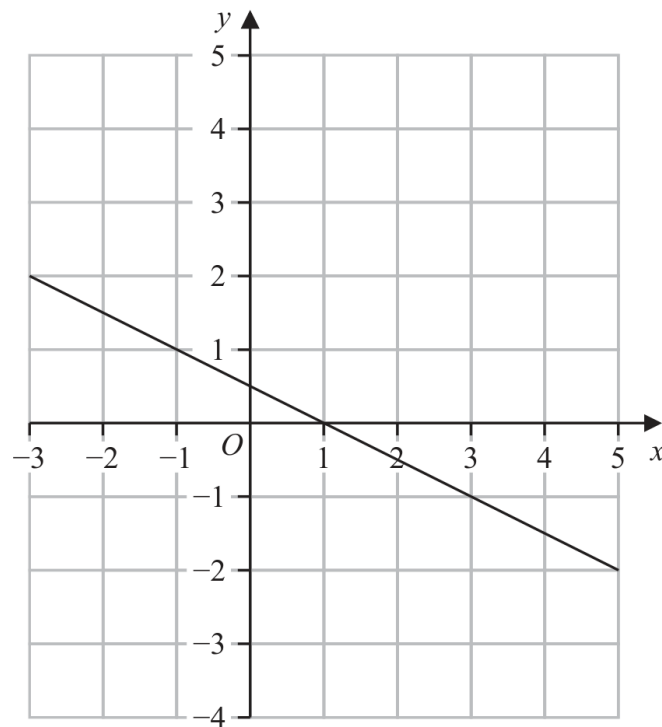
Question 29

4MA1-2H Q14 whole

MODULAR UNIT 2

3 marks

14 Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$

$$2y + x = 1$$

$x = \dots\dots\dots$

$y = \dots\dots\dots$

Question 30

4MA1-2H Q18 whole

MODULAR UNIT 2

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

WORKING SPACE

Question 31

4MA1-2H Q19 (a)

MODULAR UNIT 2

1 marks

19 The functions f and g are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x-9}$$

(a) State which value of x cannot be included in any domain of g

.....
(1)

WORKING SPACE

Question 32

4MA1-2H Q19 (b)

MODULAR UNIT 2

2 marks

19 The functions f and g are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x - 9}$$

(b) Find $fg(4)$

.....
(2)

WORKING SPACE

Question 33

4MA1-2H Q19 (c)

MODULAR UNIT 2

4 marks

The function h is such that

$$h:x \mapsto 3x^2 - 12x + 8 \quad \text{where } x > 2$$

(c) Express the inverse function h^{-1} in the form $h^{-1}:x \mapsto \dots$

$$h^{-1}:x \mapsto \dots\dots\dots (4)$$

WORKING SPACE

Question 34

4MA1-2H Q21 whole

MODULAR UNIT 2

5 marks

- 21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(..... ,)

(..... ,)

WORKING SPACE

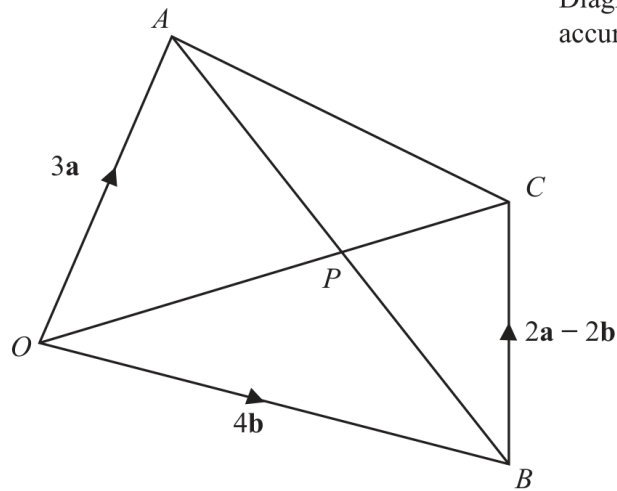
Question 35

4MA1-2H Q22 (a)(i), (a)(ii), (b)

MODULAR UNIT 2

5 marks

22

Diagram **NOT**
accurately drawn $OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots$$

(1)

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots$$

(1)

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

$$\dots\dots\dots$$

(3)

Working space for Question 35

Continue your method

UNIT 2

5 marks

WORKING SPACE

Question 36

4MA1-2H Q23 whole

MODULAR UNIT 2

5 marks

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

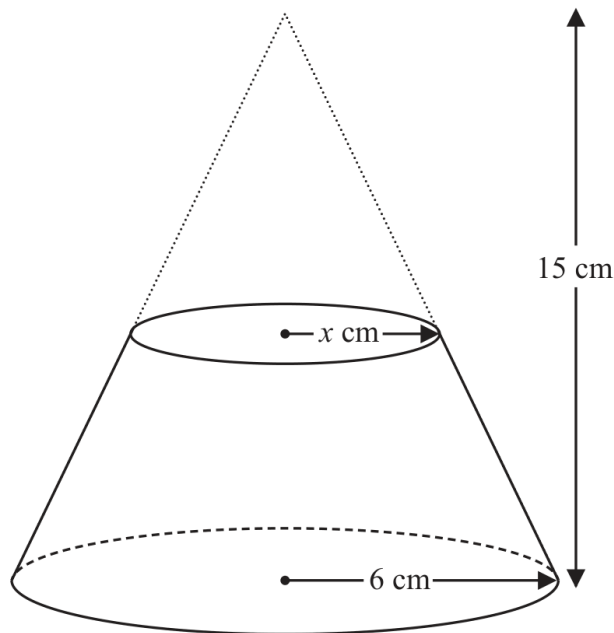


Diagram **NOT**
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$

work out the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

WORKING SPACE

Working space for Question 36

Continue your method

UNIT 2

5 marks

WORKING SPACE